imo-2024-01

Divisibility of floor sums

Arithmétique / Number theory | Level 5/5 (provisional)

Which real values of α satisfy

$$n \mid \sum_{k=1}^{n} \lfloor k\alpha \rfloor \qquad \text{for every integer } n \geq 1?$$

Here $\lfloor t \rfloor$ means the largest integer not exceeding t.

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Editorial restatement of IMO 2024 problem 1; mathematical conditions preserved, not official wording.

Statement checked against source by Codex. Independent human review pending.